Foundations of Computer Programming: Operators and Expressions

Learning objectives:

- Understand the Purpose of Expressions in Python
- Apply Arithmetic and Assignment Operators
- Use Comparison and Logical Operators to Make Decisions
- Understand and Use Bitwise Operators
- Work with Membership and Identity Operators
- Demonstrate Operator Precedence and Associativity
- Construct and Evaluate Python Expressions

Lesson exercise: Week 8

1. Provide a python example of following arithmetic operators:

Operator
+
-
*
/
%
**
//

2. Provide a python example of following assignment operators:

Operator
=
+=
-=
*=
/=
%=
//=

**=
&=
|=
^=
>>=
<<=
:=

3. Provide a python example of following comparison operators

Operator
==
!=
>
<
>=
<=

4. Provide a python example of following logical operators

Operator
and

or
not

5. Provide a python example of following identity operators

Operator
is
is not

6. Provide a python example of following membership operators

Operator
in
not in

7. Provide a python example of following bitwise operators

Operator
&
|
^
~

<<
>>

8. Assign the value 10 to a variable a.

   Use arithmetic and assignment operators to:

   Add 5 to a.

   Subtract 3 from a.

   Multiply a by 2.

   Divide a by 4 (use floor division).

   Print the result of a.

9. Define two variables, x = 7 and y = 12.

   Use comparison operators to check if:

   x is greater than y

   x is less than or equal to y

   Combine these comparisons using logical operators to:

   Check if x is less than y and x is not equal to y

   Check if x is greater than y or y is greater than 10

   Print the results of each check.

10. Define a list my_list = [1, 2, 3, 4, 5].

    Check if:

    The number 3 is in my_list.

    The number 10 is not in my_list.

Define two variables, a = [1, 2] and b = [1, 2].

Check if:

a is b

a == b